EXTENSION TO THE PIPELINE BUILD SPEC · SIGNAL 01

Job Signal Acquisition

The boards everyone names are the wrong list for this ICP. Small law firms largely avoid the expensive ones — which means the sources with the best coverage of our targets are also the free, structured, publicly crawlable ones.

Lanes 3 **Components** JA1–JA6 **Hard problems** 5

§1 Where our firms actually post

The obvious answer to "which job boards matter" is Indeed, LinkedIn, ZipRecruiter. For general hiring that is right. For *2–25 attorney law firms hiring support staff*, it is misleading — and the difference decides the whole architecture.

Posting a role on the large boards costs money per listing or per click. A 4-attorney firm filling one paralegal seat does not run a recruiting budget, so it posts where posting is free or already paid for: **its own website, its bar association, and the niche legal boards it already belongs to.**

That reverses the usual trade-off. Normally the compliant sources are the weak ones and the good data sits behind terms that prohibit collecting it. Here the sources with the *best coverage of our exact segment* are also the ones that publish structured, machine-readable postings and invite crawling.

> WHAT FOLLOWS FROM THIS
>
> Build lane 1 — own careers pages, bar boards, ATS endpoints — first and completely. It covers the ICP better than the big boards do, it is free, it is stable, and nothing about it breaks when a vendor changes its terms.
>
> The large boards are a **coverage top-up**, not the foundation. They matter for the larger end of the range and for firms that do run a recruiting budget, and they are bought rather than scraped.

§2 Board register

Every meaningful US source for legal support-staff postings, with the access route that actually works in 2026 rather than the one that used to.

TIER A — BEST COVERAGE OF OUR ICP, PUBLICLY ACCESSIBLE

SOURCE	WHY IT MATTERS HERE	ACCESS ROUTE	VERDICT
The firm's own careers page	Where small firms post first and often only. Also the one place the posting is unambiguously tied to the firm — no name resolution needed.	Crawl. `JobPosting` JSON-LD is the standard markup, so most pages parse deterministically.	PRIMARY
State bar association job boards 50 states, most with a career centre	Free or cheap for members, so firms use them heavily. High legal purity — almost no noise from other industries.	Public listings crawl. A minority sit behind member login — those are out of scope, not a target.	PRIMARY
Metro & county bar associations LA County, Chicago, Houston, Dallas, Miami-Dade, Brooklyn, and so on	Even closer to the target. A county bar board is where a 6-attorney PI firm posts a paralegal role.	Public listings crawl. Long tail — prioritise the metros in the collection plan.	PRIMARY
Paralegal association boards NALA Career Center, NFPA, NALS, state paralegal associations	Role-specific and precisely on target — these boards carry almost nothing but the roles we sell.	Public listings crawl.	PRIMARY
ATS-hosted boards Greenhouse, Lever, Workable, JazzHR, BambooHR, Recruitee	Firms at the larger end of the range run an ATS. Small share of our universe, but perfectly clean data where present.	Public per-company JSON endpoints — e.g. `boards-api.greenhouse.io/v1/boards/{co}/jobs`, `api.lever.co/v0/postings/{co}`. Structured, free, documented.	PRIMARY

TIER B — HIGH VOLUME, LICENSED ACCESS

SOURCE	WHY IT MATTERS HERE	ACCESS ROUTE	VERDICT
Indeed	Largest single volume of US support-staff postings. Skews to firms that do pay to advertise roles.	No public API — the Publisher API was retired and the remaining Apply API is partner-only. Route is a **licensed job-data provider** (Coresignal, Bright Data, Techmap, TheirStack and similar), which returns normalised JSON.	BUY, DON'T SCRAPE
LinkedIn Jobs	Attorney and administrator roles more than support staff. Useful for the lateral-hire signal.	Licensed provider only. Direct automated collection carries account bans and real legal exposure — it is not a technical obstacle to route around.	BUY, DON'T SCRAPE
ZipRecruiter	SMB-heavy, which fits the segment better than LinkedIn does.	Has a partner API programme. Apply for it; otherwise reach it through the same licensed providers.	PARTNER API
Craigslist metro jobs → legal/paralegal	Genuinely still used by small firms for admin and support roles, especially in large metros. Consistently underestimated.	Public RSS per city/category exists. Respect rate limits and terms; treat as a low-volume supplementary feed.	LIGHT TOUCH

SOURCE	WHY IT MATTERS HERE	ACCESS ROUTE	VERDICT
Glassdoor · SimplyHired	Largely mirror Indeed's index. Little independent signal.	Via the same licensed providers if wanted. Low marginal value.	SKIP INITIALLY

TIER C — LEGAL NICHE AGGREGATORS

SOURCE	NOTE	ACCESS ROUTE	VERDICT
LawCrossing	Aggregates legal jobs including from firm sites. Subscription product — and its own listings are partly re-published from sources we already cover directly.	Subscription. Evaluate for overlap before paying.	EVALUATE
Lawjobs.com · Law.com ALM	Skews larger firms and attorney roles.	Commercial. Low fit with the 2–25 band.	LOW PRIORITY
ABA Career Center	National, attorney-weighted, moderate support-staff volume.	Public listings.	SUPPLEMENTARY
Vault · Leopard Solutions	Large-firm intelligence products. Our disqualified band.	—	OUT OF SCOPE
Law school career portals Symplicity and similar	Authenticated, institution-scoped.	Requires credentials we have no basis to hold.	EXCLUDED

TIER D — THE QUERY LANE

SOURCE	NOTE	ACCESS ROUTE	VERDICT
Google for Jobs the aggregated jobs panel	Not a board — an index built from `JobPosting` markup across the web. Its value is *discovery*: it finds postings on sites we have not thought to crawl.	Google retired its Jobs API, so there is no endpoint to query directly. Reach it through a SERP data provider — DataForSEO, which the stack already uses, or SerpApi.	DISCOVERY LANE

§3 Three lanes, one signal table

Three acquisition modes with different economics and different failure modes, converging on one deduplicated signal. Each lane can fail without taking the others down.

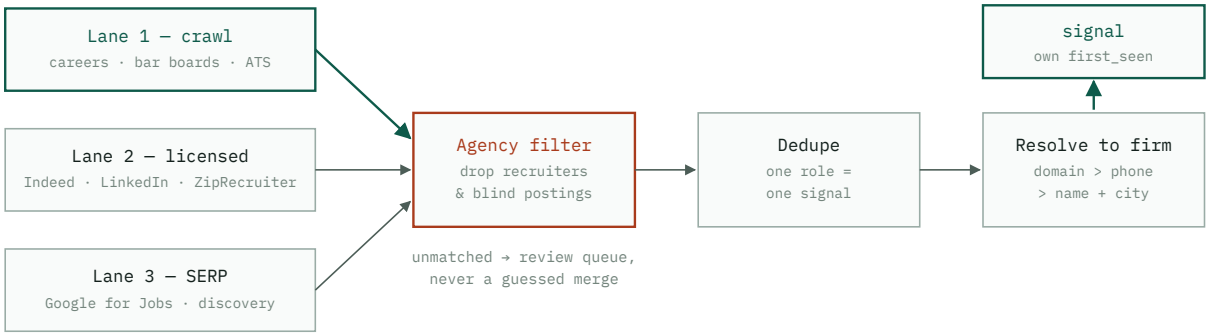


Fig 1 — The agency filter sits before deduplication deliberately: a staffing agency posting the same role in eleven cities would otherwise survive as eleven distinct signals against eleven wrong firms. Nothing writes a signal until it has resolved to a real firm.

§4 The agency filter

A large share of legal support-staff postings are placed by staffing and recruiting agencies, not by law firms. These are not leads — they are a different industry, and the firm that actually needs the person is hidden behind "our client".

Without this filter the pipeline targets recruiters as if they were prospects, inflates every volume estimate, and writes signals onto companies that will never buy. It is the single highest-value transform in this whole lane.

```
-- reject a posting when ANY of these fire

1. KNOWN AGENCY
   company matches the maintained legal-staffing blocklist
   -- Robert Half Legal, Beacon Hill Legal, Adams & Martin,
   -- Special Counsel / Adecco, Vaco, Lateral Link, Kforce Legal,
   -- Marina Sirras, Latitude, Hire Counsel … maintained, not static

2. CLIENT LANGUAGE
   body matches /\bour client\b|\bconfidential client\b/i
             |  /\b(?:a|our)\s+(?:growing|established|boutique)\s+
                  (?:law\s+firm|firm)\b/i   -- describes rather than names

3. GEOGRAPHIC SPREAD   -- the strongest behavioural tell
   same company posting legal support roles
   in >= 5 distinct metros within 90 days
   -- a real 2–25 attorney firm does not hire paralegals in
   -- five cities at once. an agency does it every week.

4. VOLUME
   same company with >= 20 open legal support roles
   -- inconsistent with the entire target size band

5. BLIND POSTING
   employer name absent, generic, or "confidential"
   -- unresolvable. drop rather than guess.

-- what happens to rejects
   written to job_posting_rejected with the rule that fired.
   NOT silently discarded — the reject stream is how the
   blocklist gets extended, and a sudden spike in rule 3
   usually means a new agency has appeared, not that the
   market changed.
```

RULE 3 CATCHES WHAT A BLOCKLIST CANNOT

Agency names change, new ones appear constantly, and no maintained list stays complete. Geographic spread is a behavioural signature rather than an identity, so it catches agencies nobody has heard of yet — and it costs one grouped query.

§5 Cross-board deduplication

One paralegal opening can appear on the firm's careers page, its state bar board, a county bar board, Indeed, ZipRecruiter and Glassdoor. That is one signal, not six — and counted six times it would push a firm's intent score to the top of the queue for no reason.

```
-- posting fingerprint: same role, same firm, same window

  fingerprint = sha1(
      firm_id                              -- after resolution
    ‖ normalize_title(title)               -- see below
    ‖ metro_code                           -- CBSA, not raw city string
    ‖ floor_to_week(earliest_posted_date)
  )

  normalize_title(t):
    lowercase; strip seniority (sr|jr|senior|junior|lead|i|ii|iii)
    strip employment terms (full[- ]time|part[- ]time|contract|temp)
    strip location fragments and firm name
    map synonyms → canonical role:
      "legal assistant"|"legal secretary"|"admin assistant, legal"
                                           → legal_assistant
      "paralegal"|"certified paralegal"|"litigation paralegal"
                                           → paralegal
      "intake specialist"|"client intake"|"intake coordinator"
                                           → intake_specialist
      "case manager"|"case coordinator"    → case_manager
      "records clerk"|"medical records"    → document_collector

-- merge policy on collision
  keep ONE signal row.
  first_seen  = MIN across all sources   -- earliest sighting wins
  last_seen   = MAX across all sources
  sources[]   = append every board it was seen on
  canonical_url = prefer the firm's own careers page,
                  then the bar board, then a licensed source

-- sources[] is not bookkeeping. a role posted on five boards is a
-- firm trying harder than one posted on a single board — that is a
-- usable intensity measure, and it only exists if we keep the list.
```

BOARD COUNT AS INTENSITY

A firm that posted one paralegal role to five different boards is spending effort and money to fill it. A firm that posted the same role only to its own careers page may be casually looking. Same signal kind, materially different urgency — and it costs nothing to capture, because deduplication already touched every copy.

§6 Resolving a posting to a firm

Lane 1 mostly solves this for free — a posting on a firm's own careers page is already attached to a domain. Lanes 2 and 3 give a company name and a city, which is the weakest matching position in the system.

The practical unlock is to **keep the full posting body, not just the structured fields.** Application emails, "visit our website at", and apply-links back to the firm's own site appear constantly in the text and each one carries a domain, which turns a probabilistic name match into a decisive one.

RESOLUTION LADDER FOR BOARD-SOURCED POSTINGS

#	EXTRACT	MATCH RULE	CONFIDENCE
1	**Apply-email domain in the body** `careers@smithlaw.com`	eTLD+1 exact against `firm.domain`	DECISIVE
2	**Outbound link to a non-board domain**	eTLD+1 exact, board and ATS hosts excluded	DECISIVE
3	**Phone number in the body**	E.164 exact	DECISIVE
4	**ATS slug** `boards.greenhouse.io/smithlaw`	Slug to firm map, built once and reused	DECISIVE
5	**Company name + street address**	Address hash plus name similarity	STRONG
6	**Company name + metro**	Trigram within CBSA. Accept only if exactly one candidate clears the threshold.	REVIEW QUEUE
7	**Company name alone**	—	REJECT Writes a false hiring signal onto a real firm. Never guess.

> AN UNMATCHED POSTING IS STILL WORTH KEEPING
>
> If a posting cannot be resolved, it may name a firm that is **not yet in the universe at all** — a discovery lead rather than a failure. Route it to the discovery queue with its company name and metro so the normal harvest can try to find that firm properly, then re-attempt the match. Do not merge it into an existing record on a hunch.

§7 Dates, and why reposts matter more than dates

Board-reported dates cannot be trusted. Many boards show relative ages that round badly — "30+ days ago" covers everything from a month to a year. Worse, employers refresh listings to keep them near the top, which resets the displayed date while the underlying vacancy is unchanged.

So the system records **its own** `first_seen`, keyed on the fingerprint from §5. That is what makes "open more than 21 days" computable at all.

```
-- three date fields, three different meanings

  board_posted_date    what the board claims        -- untrusted, stored
  first_seen           when WE first saw this
                       fingerprint                  -- the operative one
  last_seen            most recent sighting         -- drives closure, §8


-- REPOST DETECTION — the strongest variant of signal 01

  same fingerprint, but board_posted_date has moved forward
  while our first_seen stayed put
    → the firm re-listed a vacancy it has not filled
    → this is an active, dated admission of hiring failure

  repost_count >= 1                  weight 44   -- above the 40
                                                 -- for "open >21d"
  repost_count >= 2                  weight 48

-- "open more than 21 days" is inferred from absence of an update.
-- a repost is the firm affirmatively spending effort again on the
-- same unfilled seat. it is a stronger claim, it is dated, and it
-- is only detectable if we hold our own first_seen against the
-- board's shifting one.
```

> THE LINE A REPOST WRITES
>
> *"Saw the paralegal role at [Firm] go back up on [board] this week — second time since [month]."* That is specific, verifiable, and impossible to write without holding the history. It is the clearest example of why keeping our own dates matters more than reading theirs.

§8 Closure detection

A signal must end when the role is filled, or the pipeline keeps working leads whose reason for existing has gone. But absence from one board is ambiguous — listings expire on schedules that have nothing to do with the vacancy.

```
-- absence on one board ≠ closed

  a posting is CLOSED only when it is absent from
  every source in sources[] across two consecutive sweeps

  precedence: the firm's own careers page is authoritative.
      gone from Indeed, still on careers page   → still OPEN
      gone from careers page, still on Indeed   → likely CLOSED
                                                  (board listing is stale)

  grace: 2 sweeps, not 1 — one failed fetch must never
         close a live signal

  on close: signal.closed_at set → weight drops to 0 → intent
            recomputes → lead may fall out of the armed band
            but the lead is not discarded. it returns to the
            pool with fit and contact intact (lifecycle §10),
            and the closed posting stays as fit history:
            this firm hires paralegals.
```

DO NOT CLOSE ON A FETCH FAILURE

A site that times out, a board that rate-limits, or a parser that breaks on a redesign all produce the same observable as a removed posting: nothing found. Distinguish *fetched successfully and absent* from *fetch failed*, and only the first counts toward closure. Without that distinction, one bad night silently closes thousands of live signals.

§9 Crawl mechanics for lane 1

PARSE ORDER — cheapest and most reliable first

1. JobPosting JSON-LD in the page — deterministic, most sites
2. microdata / RDFa — older sites
3. ATS JSON endpoint if the page embeds one — fully structured
4. HTML parse with per-template selectors — bar boards: ~40 templates
5. LLM extraction, low volume only — last resort, flagged

BAR BOARDS

~50 state + ~60 metro boards, but only about 40 distinct platforms behind them (YourMembership, Naylor, MemberClicks, Careerwebsite, Personify …). Write parsers per **platform**, not per board — coverage per parser is roughly 3:1.

Member-login boards are out of scope. Not an obstacle to route around; simply not a source.

RATE DISCIPLINE

per-host token bucket, ≤ 1 request / 2s, honest User-Agent with a contact address, robots.txt honoured, Crawl-delay respected where declared.

-- these are small association servers. being a good citizen
-- here is also what keeps the source working long-term.

CADENCE

careers pages	nightly, all qualified firms
bar boards	nightly, metros first
ATS endpoints	nightly, cheap
licensed feed	daily delta pull
SERP discovery	weekly, rotating query set

CHANGE DETECTION

store content hash per listing page.
unchanged hash → skip parse entirely.

-- most nights, most pages are unchanged. this turns a full
-- sweep into a small one and is what makes nightly feasible.

§10 Components

Six components. JA1–JA3 replace and expand the placeholder watchers W1 and W2 in the build spec.

JOB ACQUISITION SUB-PLANE

ID	COMPONENT	INPUT → OUTPUT	CADENCE	ON FAILURE
JA1	**Careers page watcher**	`firm.domain` → `raw_posting`	Nightly	Per-firm isolation; a broken site never stalls the sweep
JA2	**Bar board watcher** parsers per platform, not per board	`board registry` → `raw_posting`	Nightly	Per-board isolation; alert when a platform parser yields zero across all its boards
JA3	**ATS endpoint poller**	`ats slug map` → `raw_posting`	Nightly	Cheap and stable; low alert priority
JA4	**Licensed feed ingester**	`provider delta` → `raw_posting`	Daily	Vendor-agnostic adapter — the provider will change; the schema should not
JA5	**SERP discovery**	`query set` → `raw_posting` + discovery queue	Weekly	Best-effort. Its job is finding sources we do not know about.
JA6	**Normaliser** agency filter → dedupe → resolve → signal	`raw_posting` → `signal`	After each sweep	Pure function over stored raw postings — fully re-runnable without re-fetching

> WHY JA6 IS SEPARATE
>
> The agency blocklist will be extended, the title synonym map will be corrected, and the resolution rules will be tuned — repeatedly, in the first months. Keeping normalisation as a pure function over stored `raw_posting` rows means every one of those changes is a re-run over what is already on disk, not a re-crawl of the internet.

§11 Build order

#	BUILD	WHY HERE
1	**JA1 careers page watcher**	Best ICP coverage, zero resolution problem, free, stable. Also the source of truth for closure detection in §8, so everything else leans on it.
2	**JA6 normaliser — agency filter first**	Even with one source, agency noise must be filtered before signals are written. Build the filter before the second source multiplies the noise.
3	**JA2 bar boards — top 10 metros**	Highest-value addition after careers pages, and the parsers are reusable across roughly three boards each. Prove the platform-parser approach on ten before writing forty.
4	**JA3 ATS endpoints**	Small effort, clean data, no parser maintenance.
5	**JA5 SERP discovery**	Reveals which sources are being missed — and that answer should inform whether lane 2 is worth buying.
6	**JA4 licensed feed**	Last. It costs money, and by this point there is a measured coverage gap to size it against instead of an assumption.

> BUY THE FEED LAST, NOT FIRST
>
> The instinct is to start with a licensed provider because it arrives as clean JSON. But until lane 1 is running there is no way to know what it adds — and given that small firms largely avoid the paid boards, the honest expectation is that its incremental coverage of *this* ICP is smaller than it looks.
>
> Run lane 1, measure what share of firms produce postings there, then buy against a number.

Extension to the Pipeline Build Spec. Expands signal 01 acquisition into six components and replaces the placeholder W1/W2 watchers.

Access routes reflect the position as of September 2026 — Indeed's Publisher API is retired with no public replacement, and Google's Jobs API has been phased out. Verify provider terms before contracting. Weights and thresholds are seeded values to be replaced by measurement.